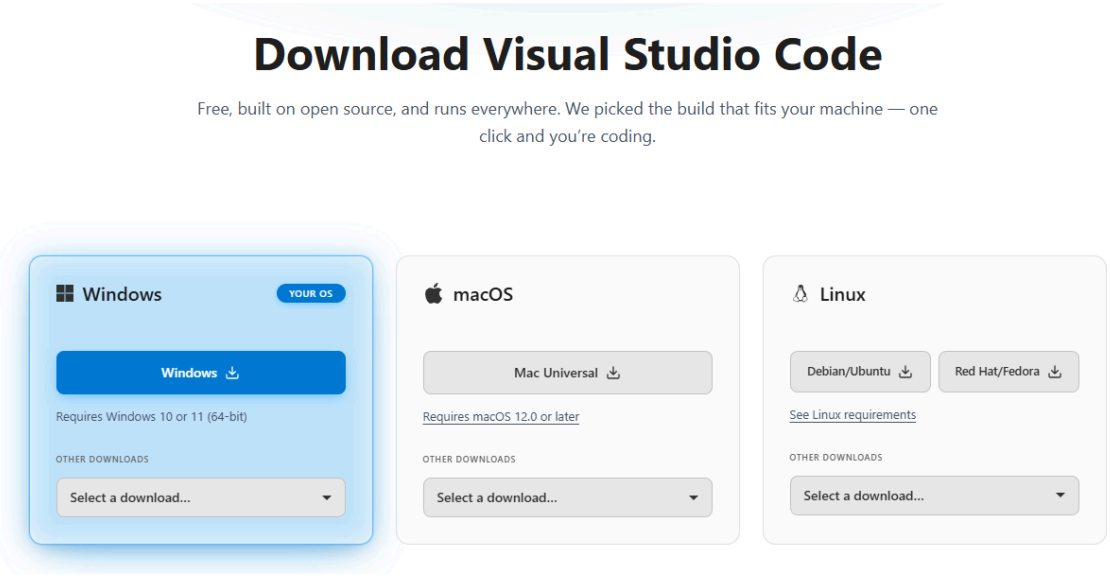
Setting Up Your Microcontroller Workspace

Prepared by Lior Brown

Goal: Install Visual Studio Code (VS Code) with PlatformIO and Wokwi, and create a ready-to-use embedded workspace for the course.

Step 1 - Download Visual Studio Code (VS Code)

1. Visit https://code.visualstudio.com/download



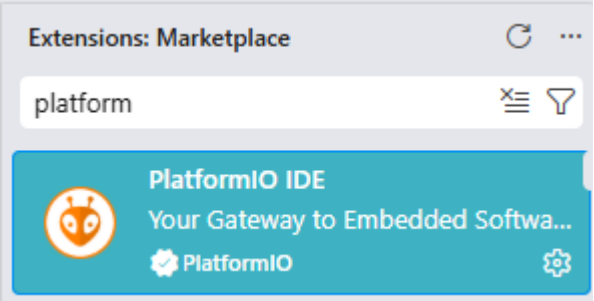
2. Select your operating system (Windows, macOS, or Linux).
3. Run the installer → default settings are fine (just keep clicking Next).

Step 2 - Install the Required Extensions

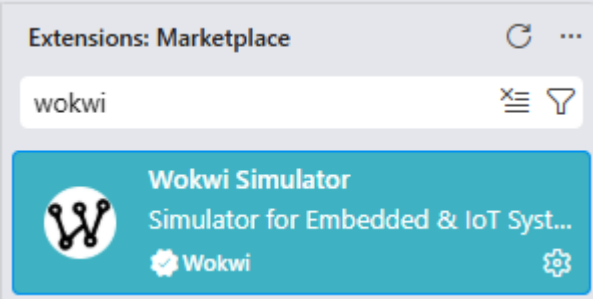
(VS Code needs specific add-ons to build and simulate microcontroller code)

1. Open VS Code.

2. Click on the **Extensions** icon  on the far left menu.
3. In the search bar, type **PlatformIO IDE**.
4. Click **Install** on the one with the ant ant logo.



5. Next, clear the search bar and type **Wokwi Simulator**.
6. Click **Install** on the extension published by *Wokwi*.

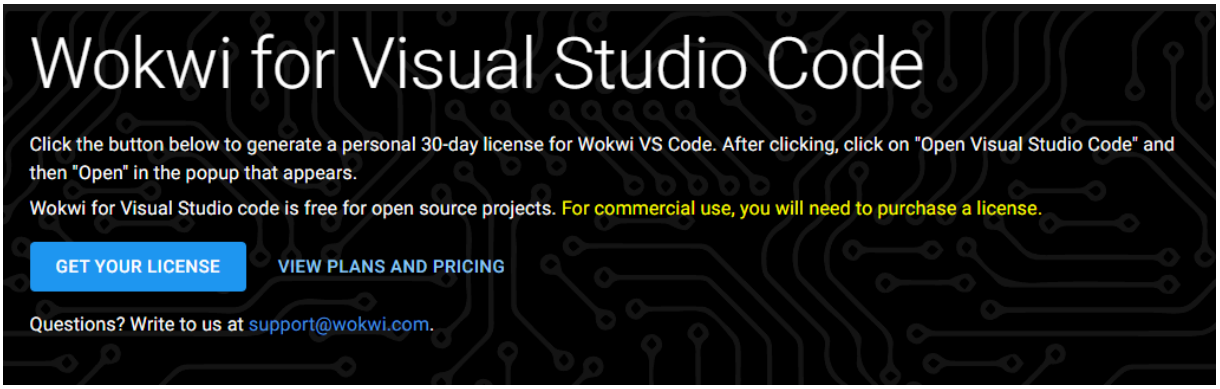


Note: PlatformIO might take a few minutes to install its core dependencies in the background. Please be patient and wait for the "Installation has been successfully finished" message in the bottom right corner.

Step 3 - Activate Your Free Wokwi License

(The Wokwi simulator will not run without a license. The free Community license is enough for this course.)

1. In VS Code, press **F1** (or **Ctrl+Shift+P**) to open the Command Palette.
2. Type **Wokwi: Request a New License** and press Enter.
3. Your browser will open the Wokwi website. Sign in (a Google or GitHub account works).
4. Click **Get Your License**.

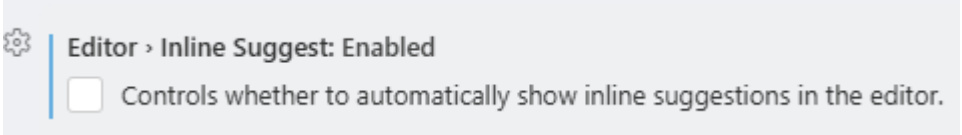


5. When the browser asks to open VS Code, allow it. You should see a message in VS Code confirming that the license was activated.

Note: The free license is valid for 30 days. When it expires, repeat this step to renew it for free.

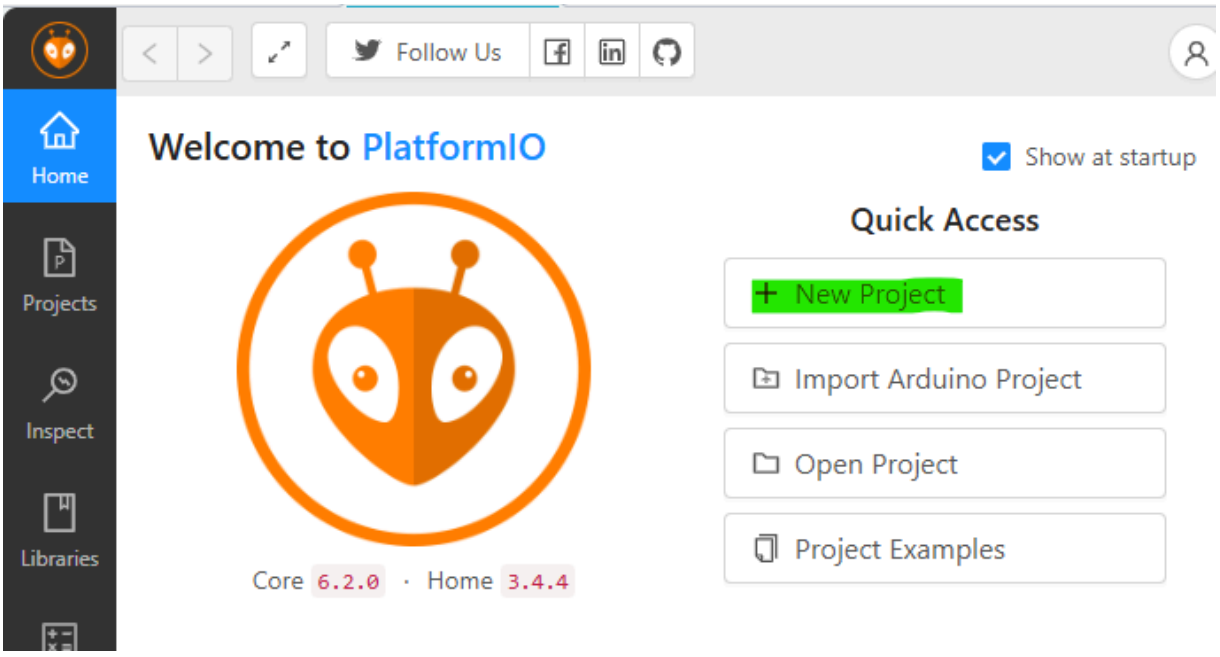
Step 4 - Disable AI Suggestions

1. Open Settings (File → Preferences → Settings, or click the Gear icon ⚙ in the bottom left).
2. In the search bar at the top, type `Inline Suggest`.
3. Uncheck the box for **Editor: Inline Suggest: Enabled**.

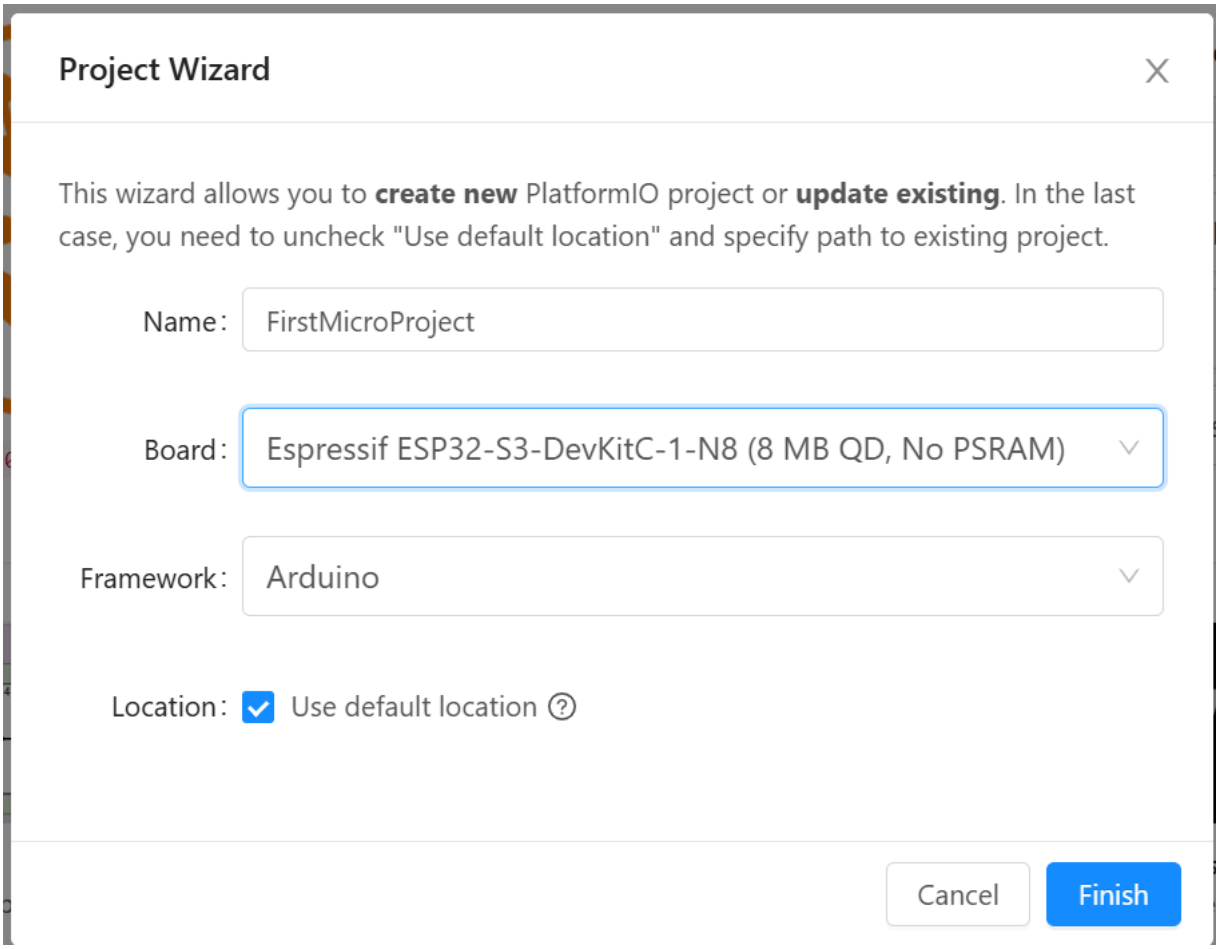


Step 5 - Create Your First Project and Configure Files

1. Click the **PlatformIO icon** (the ant head) on the left sidebar to open PIO Home.
2. Click the **New Project** button.



3. Name your project (e.g., `FirstMicroProject`), select the board as **Espressif ESP32-S3-DevKitC-1**, and ensure Framework is **Arduino**. Click Finish.
   (Note: This initial project creation may take a lot of time as PlatformIO downloads the heavy toolchains and frameworks in the background. Please be patient and let it finish!)



4. **Important Config Step:** To enable the simulator for this course, you need two configuration files that I have provided: `wokwi.toml`, and `diagram.json`.
5. Open your project folder in your computer's file explorer.
6. **Copy** the `wokwi.toml` and `diagram.json` files I provided and paste them directly into that same main project folder.
   (`wokwi.toml` tells Wokwi where to find your compiled code, and `diagram.json` tells it which board to simulate.)

Step 6 - Write and Run Your Code

1. In VS Code, open the Explorer menu on the left, expand the `src` folder, and click on `main.cpp`.
2. Replace the default code in the editor with the following:

```cpp
#include "Arduino.h"

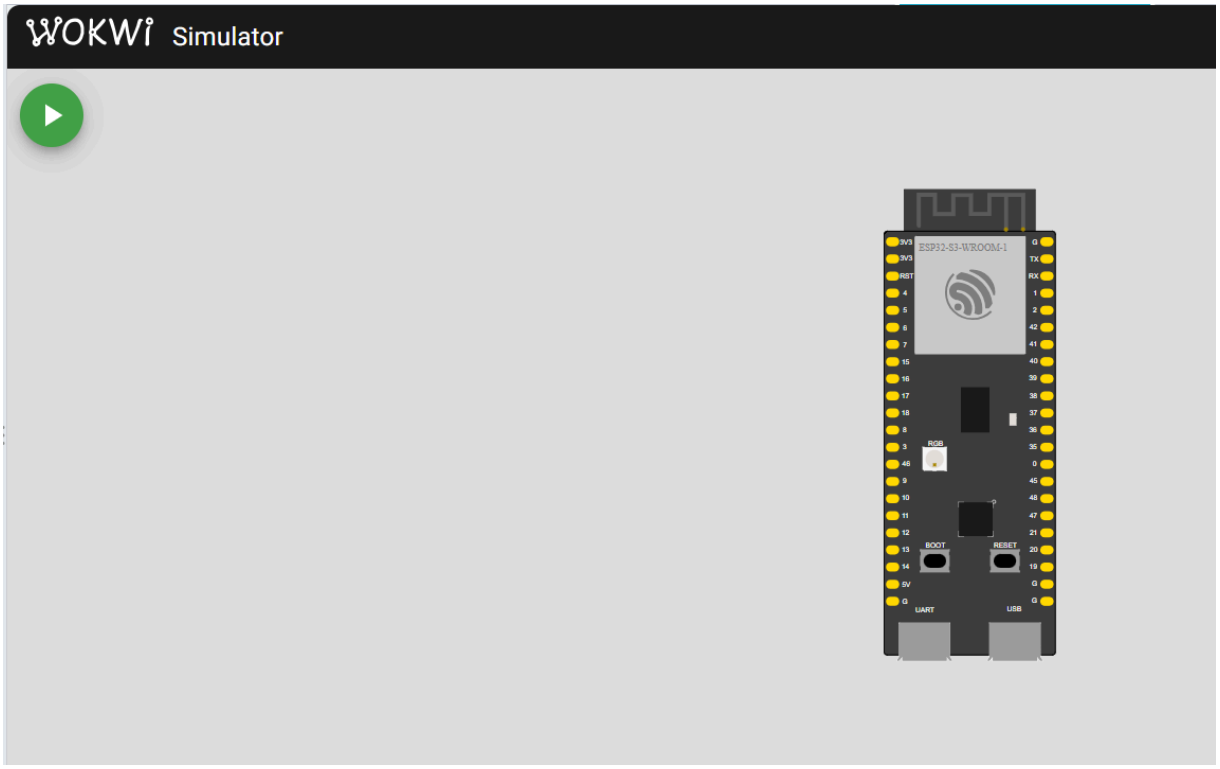
void setup() {
    Serial.begin(115200);
    Serial.println("Am Israel Chai");
}

void loop() {
    // Do nothing
}
```

3. Click the PlatformIO Build button (the checkmark ✓ icon) at the very bottom status bar to compile your code.



4. To simulate it, open the `diagram.json` file from the Explorer panel on the left, then click the green Play button (▶) in the simulator window to start the simulation.



5. Look at the Terminal panel at the bottom → you should see:

```plaintext
Am Israel Chai
```

(If you don't see it - don't worry, Am Israel is still alive 🇮🇱*)*